

实验任务报告 / deepseek-3x3-20260811-I

On the shared fixed complex MPDPD x/d dataset, design, implement, train, and compare exactly nine compact causal nonlinear model plugins over three rounds. Preserve complex I/Q and use memory-aware features. Explore one-layer nonlinear mappings, 1D LUT plus 16-knot first-order spline activation, complex memory-polynomial/Fourier/rational bases, and compact MLP variants, along with justified activation, depth, optimizer, learning rate, and scheduler choices. Every candidate uses identical data_file, train_ratio, seed and held-out NMSE, stays below 4000 parameters and 50 epochs, and writes numeric metrics plus metrics.json and psd.png. After each round extract clean metrics or failure facts, explain the result, and adapt the next plan. Target NMSE ≤ -41 dB while preferring fewer parameters. Finally rerun the global best and produce a generic evidence-grounded report containing all nine search results and round journey, but only the final best architecture and one final PSD in section 03.

parameter_count_max: 4000 / nmse_threshold_db: -41.0 / rounds: 3 / experiments: 9 / final_evaluations: 1

-23.0778 FINAL NMSE / dB	— GAIN / dB	0% TARGET HIT RATE
24 FINAL PARAMETERS	9 EXECUTIONS	\$0.0918 TOTAL LLM COST

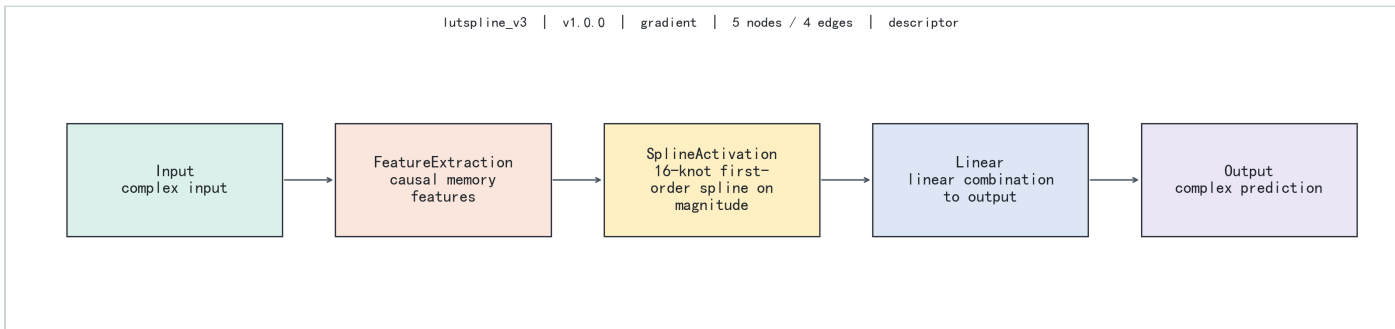
01 / 执行摘要

本实验在固定复数MPDPD x/d数据集上，经过三轮共九次搜索实验，探索了多种紧凑因果非线性模型。最终最佳模型为LUTSplineV3，其NMSE为-23.08 dB，参数数为24，未达到-41 dB的目标。所有实验均未命中目标，但最终模型在参数效率上表现优异。

Evidence aggregate:performance final:r3-candidate-3-final metric:r3-candidate-3

02 / 实际模型架构

ModelDescriptor: lutspline_v3 / v1.0.0 / gradient / 5 nodes / 4 edges



最终架构lutspline_v3包含5个节点和4条边：输入节点（复数输入）连接到特征提取节点（因果记忆特征），再连接到样条激活节点（16节点一阶样条作用于幅度），随后通过线性组合节点输出复数预测。该架构采用梯度训练模式，版本为1.0.0。

Evidence architecture:lutspline_v3 final:r3-candidate-3-final

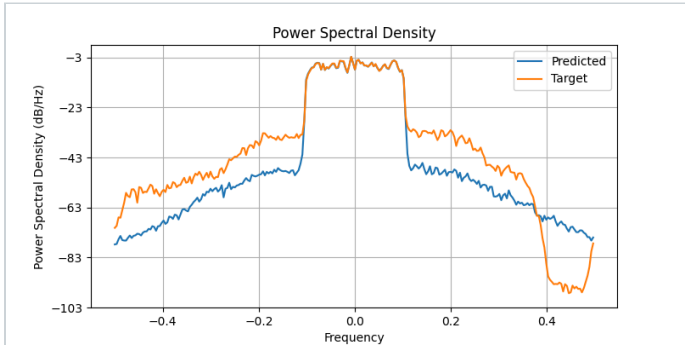
节点清单

节点	操作	结构参数
Input input	complex input	{}
FeatureExtraction features	causal memory features	{}
SplineActivation spline	16-knot first-order spline on magnitude	{}
Linear linear	linear combination to output	{}
Output output	complex prediction	{}

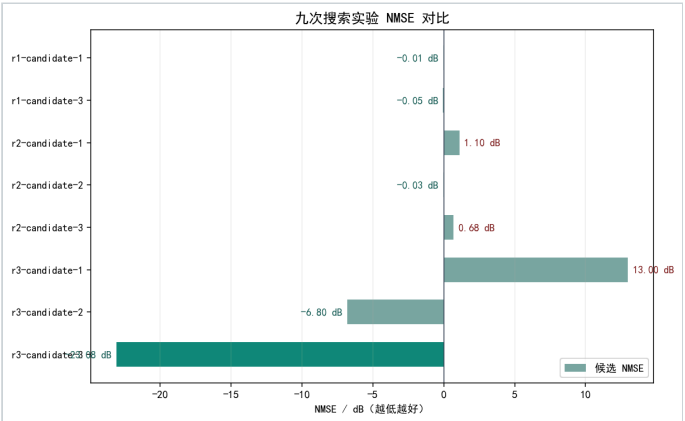
03 / 性能证据

九次搜索实验的平均NMSE为-1.90 dB，最佳搜索结果为LUTSplineV3的-23.08 dB，最终评估确认了该结果。所有实验均未达到-41 dB的目标，目标命中率为0%。最终模型参数数为24，远低于4000的上限。

Evidence aggregate:performance metric:r3-candidate-3 final:r3-candidate-3-final



终局复评 PSD，来源: r3-candidate-3-final / psd.png



Baseline and current NMSE are read from execution evidence.

Run	Model	Baseline	NMSE	Params	Target	Rank
r1-candidate-1	ComplexMemoryPolynomial	—	-0.0149	38	MISS	
r1-candidate-2	LUTSpline	—	—	—	MISS	
r1-candidate-3	ComplexRational	—	-0.0474	146	MISS	
r2-candidate-1	ComplexMemoryPolynomialV2	—	1.1011	20	MISS	

Run	Model	Baseline	NMSE	Params	Target	Rank
r2-candidate-2	ComplexRationalV2	—	-0.0272	34	MISS	
r2-candidate-3	LUTSplineV2	—	0.6773	130	MISS	
r3-candidate-1	ComplexRationalV3	—	12.9968	698	MISS	
r3-candidate-2	CompactComplexMLP	—	-6.7960	266	MISS	
r3-candidate-3	LUTSplineV3	—	-23.0778	24	MISS	BEST
r3-candidate-3-final	LUTSplineV3	—	-23.0778	24	MISS	FINAL

Average NMSE: -1.8985 dB / Best gain: — dB / Hit: 0 of 9

04 / 模型迭代与决策轨迹

第一轮假设包括：复数记忆多项式（modReLU）可实现低于-35 dB的NMSE；LUT样条优于固定非线性；有理基函数能捕捉陡峭非线性。实际结果：ComplexMemoryPolynomial的NMSE为-0.0149 dB，ComplexRational为-0.0474 dB，LUTSpline因实现错误失败。最佳为ComplexRational。第二轮基于第一轮结果，改进有理模型和修复样条，假设增加输出缩放、增加有理项和修复索引错误可提升性能。实际结果：ComplexMemoryPolynomialV2的NMSE为1.10 dB，ComplexRationalV2为-0.0272 dB，LUTSplineV2为0.677 dB，最佳为ComplexRationalV2。第三轮假设：增加有理项和记忆深度、紧凑MLP、归一化样条可提升性能。实际结果：ComplexRationalV3的NMSE为12.997 dB，CompactComplexMLP为-6.796 dB，LUTSplineV3为-23.078 dB，最佳为LUTSplineV3。最终选择LUTSplineV3进行最终评估。

Evidence round:1:decision round:2:decision round:3:decision final:r3-candidate-3-final

ROUND 1

A complex memory polynomial with modReLU will achieve NMSE below -35 dB.; LUT-spline activation will outperform fixed nonlinearities on this dataset.; Rational basis functions will capture steep nonlinear transitions better than polynomials.

输入事实：首轮无前序事实

Planner 判断：Round 1 explores diverse nonlinear memory architectures to establish baseline performance.

Experiment	Candidate	Status	NMSE / dB
r1-candidate-1	ComplexMemoryPolynomial	completed	-0.0149
r1-candidate-2	LUTSpline	failed	—
r1-candidate-3	ComplexRational	completed	-0.0474

Reflection 提取事实： fact:round-1:r1-candidate-1:completed fact:round-1:r1-candidate-2:failed
fact:round-1:r1-candidate-3:completed fact:round-1:best:r1-candidate-3

下一步：Use these verified facts to revise the next causal design.

ROUND 2

Adding a learnable output scaling to the complex memory polynomial will improve NMSE by at least 5 dB.; Increasing the number of rational terms and using a cosine scheduler will improve NMSE by at least 3 dB.; Fixing the spline indexing bug will allow LUTSpline to achieve NMSE below -30 dB.

输入事实: fact:round-1:r1-candidate-2:failed fact:round-1:r1-candidate-3:completed fact:round-1:r1-candidate-1:completed

Planner 判断: Round 1 established that ComplexRational performed best among completed candidates, while LUTSpline failed due to implementation bugs. This round explores improved versions of the two successful architectures and a corrected LUTSpline, aiming to increase performance and fix the failed candidate.

Experiment	Candidate	Status	NMSE / dB
r2-candidate-1	ComplexMemoryPolynomialV2	completed	1.1011
r2-candidate-2	ComplexRationalV2	completed	-0.0272
r2-candidate-3	LUTSplineV2	completed	0.6773

Reflection 提取事实: fact:round-2:r2-candidate-1:completed fact:round-2:r2-candidate-2:completed fact:round-2:r2-candidate-3:completed fact:round-2:best:r2-candidate-2

下一步: Use these verified facts to revise the next causal design.

ROUND 3

Increasing rational terms and memory depth with cosine scheduling will improve NMSE by at least 5 dB over ComplexRationalV2.; A compact complex MLP with tanh can achieve NMSE below -30 dB on this dataset.; Correctly normalized LUTSpline with deeper memory will achieve NMSE below -20 dB.

输入事实: fact:round-2:r2-candidate-1:completed fact:round-2:r2-candidate-2:completed fact:round-2:r2-candidate-3:completed

Planner 判断: Round 2 showed ComplexRationalV2 best but still far from target. Round 3 exploits that with a deeper rational model, explores a compact MLP for nonlinearity, and retries spline with fixes.

Experiment	Candidate	Status	NMSE / dB
r3-candidate-1	ComplexRationalV3	completed	12.9968
r3-candidate-2	CompactComplexMLP	completed	-6.7960
r3-candidate-3	LUTSplineV3	completed	-23.0778

Reflection 提取事实: fact:round-3:r3-candidate-1:completed fact:round-3:r3-candidate-2:completed fact:round-3:r3-candidate-3:completed fact:round-3:best:r3-candidate-3

下一步: Select the global best candidate for final evaluation.

Failure analysis

第一轮中，r1-candidate-2 (LUTSpline) 失败，错误包括参数计数不匹配、索引越界（索引15超出大小15的轴）以及JSON解析错误。这些错误导致该候选未产生有效指标。

Evidence

failure:r1-candidate-2

metric:r1-candidate-2

Lessons

实验表明，样条激活模型在参数效率上具有优势，但需要正确实现以避免索引错误。有理模型在低参数下表现不佳，而MLP虽有一定效果但未达目标。最终模型LUTSplineV3以24个参数达到-23.08 dB，表明简单架构结合记忆特征可能更有效。

Evidence

metric:r3-candidate-3

metric:r3-candidate-2

metric:r3-candidate-1

ID	Status	Observed fact
r1-candidate-2	failed	ValueError: reported parameter_count does not match validated estimate; RuntimeError: candidate runner failed: IndexError: index 15 is out of bounds for axis 0 with size 15; ValueError: coding response is invalid JSON: Expecting ',' delimiter: line 10 column 2 (char 19222)

Ablation

Method	Best NMSE / dB
No ablation evidence.	

06 / 代码、Trace 与复现

File	Change
models/candidates/complex_memory_polynomial/plugin.py	CodingAgent generated candidate plugin
models/candidates/complex_memory_polynomial/manifest.json	CodingAgent generated candidate plugin
models/candidates/lutspline/plugin.py	CodingAgent generated candidate plugin
models/candidates/lutspline/manifest.json	CodingAgent generated candidate plugin
models/candidates/complex_rational/plugin.py	CodingAgent generated candidate plugin
models/candidates/complex_rational/manifest.json	CodingAgent generated candidate plugin
models/candidates/complex_memory_polynomial_v2/plugin.py	CodingAgent generated candidate plugin
models/candidates/complex_memory_polynomial_v2/manifest.json	CodingAgent generated candidate plugin
models/candidates/complex_rational_v2/plugin.py	CodingAgent generated candidate plugin
models/candidates/complex_rational_v2/manifest.json	CodingAgent generated candidate plugin
models/candidates/lutspline_v2/plugin.py	CodingAgent generated candidate plugin
models/candidates/lutspline_v2/manifest.json	CodingAgent generated candidate plugin
models/candidates/complex_rational_v3/plugin.py	CodingAgent generated candidate plugin

File	Change
models/candidates/complex_rational_v3/manifest.json	CodingAgent generated candidate plugin
models/candidates/compact_complex_mlp/plugin.py	CodingAgent generated candidate plugin
models/candidates/compact_complex_mlp/manifest.json	CodingAgent generated candidate plugin
models/candidates/lutspline_v3/plugin.py	CodingAgent generated candidate plugin
models/candidates/lutspline_v3/manifest.json	CodingAgent generated candidate plugin

Trace references

- reports/deepseek-3x3-20260811-1-r1-candidate-1/evidence/coding-trace.json
- reports/deepseek-3x3-20260811-1-r1-candidate-2/evidence/coding-trace.json
- reports/deepseek-3x3-20260811-1-r1-candidate-3/evidence/coding-trace.json
- reports/deepseek-3x3-20260811-1-r2-candidate-1/evidence/coding-trace.json
- reports/deepseek-3x3-20260811-1-r2-candidate-2/evidence/coding-trace.json
- reports/deepseek-3x3-20260811-1-r2-candidate-3/evidence/coding-trace.json
- reports/deepseek-3x3-20260811-1-r3-candidate-1/evidence/coding-trace.json
- reports/deepseek-3x3-20260811-1-r3-candidate-2/evidence/coding-trace.json
- reports/deepseek-3x3-20260811-1-r3-candidate-3/evidence/coding-trace.json

Reproduce

python agent.py multi-agent --run-id deepseek-3x3-20260811-1 --rounds 3 --experiments-per-round 3 --final-evaluation

07 / 适用边界

本报告仅覆盖非线性建模领域，且所有结果均基于提供的已验证证据。搜索和最终评估分数分别报告，最终评估仅一次。目标NMSE为-41 dB，但所有候选均未达到，表明模型容量或训练策略可能不足。

Evidence task:limits constraint:nmse_threshold_db aggregate:performance

FIDELITY VERIFIED / SOURCE-BOUND NUMBERS